

HW1 Solutions

ECE 310: Digital Signals Processing, Spring 2025

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Solution:

a)

$$\begin{aligned} 2e^{-j\frac{\pi}{4}} + 3e^{j\frac{\pi}{4}} &= 2\left(\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}\right) + 3\left(\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}\right) \\ &= \frac{5}{\sqrt{2}} + j\frac{1}{\sqrt{2}}, \text{ Cartesian} \\ &= \sqrt{\frac{5^2}{2} + \frac{1^2}{2}} e^{j \arctan(\frac{1}{5})} \\ &= \sqrt{13} e^{j \arctan(\frac{1}{5})}, \text{ Polar} \end{aligned}$$

b)

$$\begin{aligned} \frac{(1-j)^2}{1+j} &= \frac{(\sqrt{2}e^{-j\frac{\pi}{4}})^2}{\sqrt{2}e^{j\frac{\pi}{4}}} \\ &= \frac{2e^{-j\frac{\pi}{2}}}{\sqrt{2}e^{j\frac{\pi}{4}}} \\ &= \sqrt{2}e^{-j\frac{3\pi}{4}}, \text{ Polar} \\ &= -1 - j, \text{ Cartesian} \end{aligned}$$

c)

$$\begin{aligned} 10e^{j\frac{3\pi}{4}} - 12e^{-j\frac{\pi}{3}} &= \frac{10}{\sqrt{2}}(-1 + j) - 12\left(\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) \\ &= \left(-\frac{10}{\sqrt{2}} - 6\right) + j\left(\frac{10}{\sqrt{2}} + 6\sqrt{3}\right), \text{ Cartesian} \\ &= \sqrt{\left(-\frac{10}{\sqrt{2}} - 6\right)^2 + \left(\frac{10}{\sqrt{2}} + 6\sqrt{3}\right)^2} e^{j \arctan\left(\frac{\frac{10}{\sqrt{2}} + 6\sqrt{3}}{-\frac{10}{\sqrt{2}} - 6}\right) + \pi} \\ &= \sqrt{244 + 60\sqrt{2} + 60\sqrt{6}} e^{j \arctan\left(-\frac{10+6\sqrt{6}}{10+6\sqrt{2}}\right) + \pi}, \text{ Polar} \\ &\approx 21.81e^{j2.21} \end{aligned}$$

d)

$$\begin{aligned}
 \left(\frac{4+j}{1-j} - \frac{j^2}{2-j^3}\right)\left(\frac{j}{9-j^{12}}\right) &= \left(\frac{(4+j)(1+j)}{(1-j)(1+j)} - \frac{j^2(2+j^3)}{(2-j^3)(2+j^3)}\right)\left(\frac{j(3+j^4)}{3(3-j^4)(3+j^4)}\right) \\
 &= \left(\frac{3+j^5}{2} - \frac{-6+j^4}{13}\right)\left(\frac{-4+j^3}{75}\right) \\
 &= \frac{17+j^{19}}{26} - \frac{-4+j^3}{25} \\
 &= -\frac{5}{26} - \frac{j}{26}, \text{ Cartesian} \\
 &= \frac{1}{\sqrt{26}} e^{\arctan(\frac{1}{5})-\pi}, \text{ Polar} \\
 &\approx -0.192 - j0.038 \\
 &\approx 0.196 e^{-j2.94}
 \end{aligned}$$

Note: We add π or $-\pi$ because $\arctan$ returns a value in $(-\frac{\pi}{2}, \frac{\pi}{2})$, while the angle in the polar form ranges from $-\pi$ to π (or 0 to 2π). For angles outside $(-\frac{\pi}{2}, \frac{\pi}{2})$, we determine the correct quadrant and adjust the result of $\arctan$ by adding π or $-\pi$ as needed.

Grading: 16 points

- +2 correct Polar expression each part
- +2 correct Cartesian expression each part
- Simplification to decimals is not required.

2

Solution:

$$\begin{aligned}
 2z^3 + 18 &= 0 \\
 z^3 &= -9 \\
 z^3 &= 9e^{j(\pi+2\pi k)}, \quad k \in \mathbb{Z} \\
 z &= (9e^{j(\pi+2\pi k)})^{\frac{1}{3}} \\
 z &= 9^{\frac{1}{3}} e^{j(\frac{\pi}{3} + \frac{2}{3}\pi k)}
 \end{aligned}$$

Defining any root by a function of k as $z(k) = 9^{\frac{1}{3}} e^{j(\frac{\pi}{3} + \frac{2}{3}\pi k)}$, we can see that $z(k)$ is a periodic function with period 3; in other word,

$$z(k+3) = z(k)$$

Therefore, there are three unique solutions for $k \in \{0, 1, 2\}$.

$$z(0) = 9^{\frac{1}{3}} e^{j\frac{\pi}{3}}, \quad z(1) = 9^{\frac{1}{3}} e^{j\pi} = -9^{\frac{1}{3}}, \quad z(2) = 9^{\frac{1}{3}} e^{j\frac{5\pi}{3}}.$$

$$z(3) = 9^{\frac{1}{3}} e^{j(\frac{\pi}{3} + 2\pi)} = z(0), \text{ same as } k = 0, \text{ not unique solution.}$$

Grading: 12 points

- +2 each correct solution
- +6 setup of complex exponential

3

Solution:

The magnitude of signal $H(\omega) = \sin(\omega/3)$ is easily computed as its absolute value, i.e

$$|H(\omega)| = |\sin(\omega/3)|, \quad (\text{Magnitude})$$

As $H(\omega)$ is a real value, the phase depends on the sign of $\sin(\omega/3)$, when $\sin(\omega/3) \geq 0$, the phase is 0, and when $\sin(\omega/3) < 0$ the phase is π (or $-\pi$). To see this, note that $H(\omega)$ can be computed as

$$\angle H(\omega) = \tan^{-1} \frac{\mathbb{I}\{\sin(\omega/3)\}}{\mathbb{R}\{\sin(\omega/3)\}}$$

where $\mathbb{I}\{\cdot\}$ denotes the imaginary part and $\mathbb{R}\{\cdot\}$ denotes the real part.

Since $\sin(\omega/3)$ is a real value, so $\mathbb{I}\{\sin(\omega/3)\} = 0$.

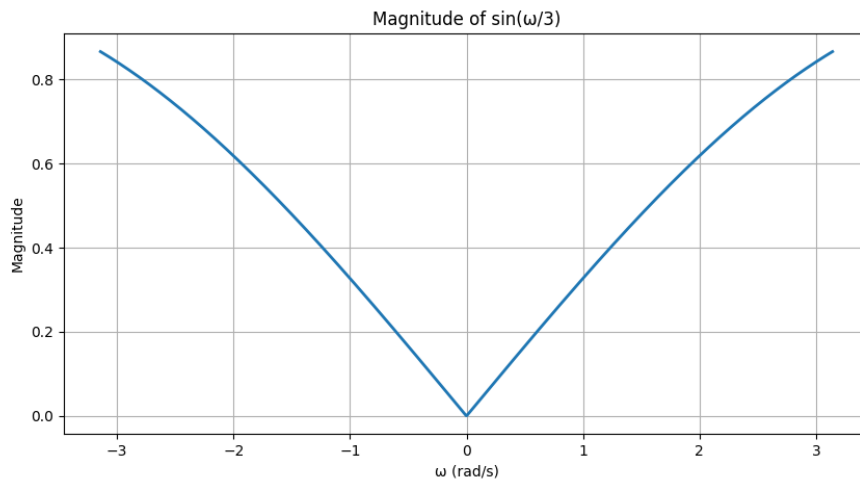
On the other hand, $\mathbb{R}\{\sin(\omega/3)\}$ is:

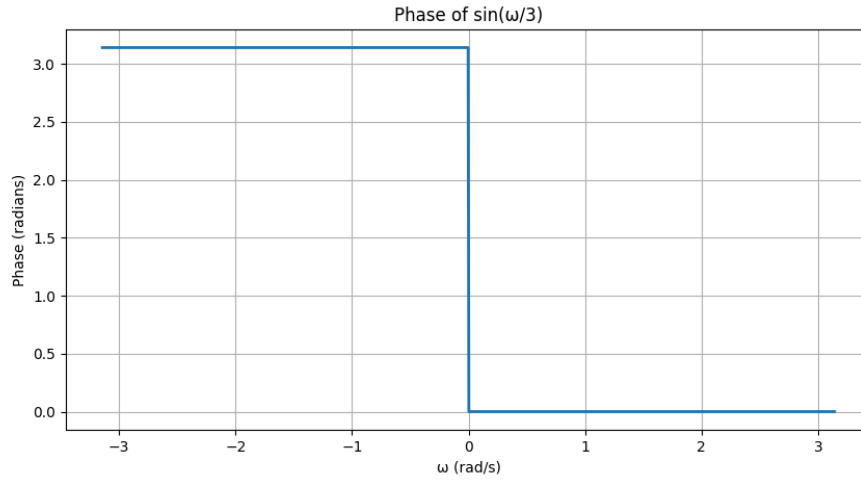
$$\mathbb{R}\{\sin(\omega/3)\} \begin{cases} < 0, & \text{for } -\pi < \omega < 0 \\ \geq 0, & \text{for } 0 \leq \omega < \pi \end{cases}$$

Therefore,

$$\angle H(\omega) = \begin{cases} \pi, & \text{for } -\pi \leq \omega < 0 \\ 0, & \text{for } 0 \leq \omega \leq \pi \end{cases} \quad (\text{Phase})$$

Plotting the magnitude and phase gives:





Grading: 12 points

- +6 each for correct magnitude/phase graph (other equivalent phase graph is fine)

4

Solution:

To find the magnitude $|H(\omega)|$, we compute the absolute value of $H(\omega)$:

$$|H(\omega)| = |\sin(\omega/3)e^{j2\omega}| = |\sin(\omega/3)| \cdot |e^{j2\omega}|$$

Since $|e^{j2\omega}| = 1$, the magnitude simplifies to:

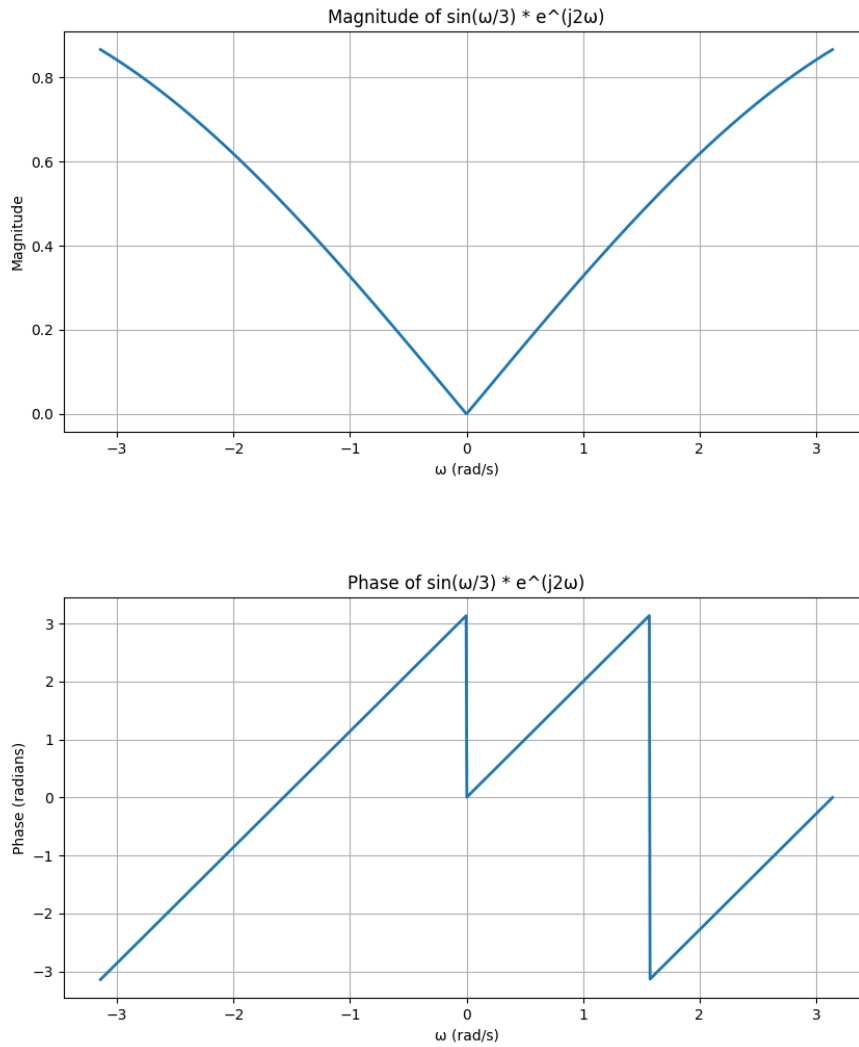
$$|H(\omega)| = |\sin(\omega/3)|, \quad (\text{Magnitude})$$

For the phase $\angle H(\omega)$, we note that $e^{j2\omega}$ contributes a phase of 2ω , hence:

$$\angle H(\omega) = \begin{cases} \pi + 2\omega, & \text{for } -\pi \leq \omega < 0 \\ 2\omega, & \text{for } 0 \leq \omega \leq \pi \end{cases} \quad (\text{Phase})$$

Since the phase of a real sine function is 0 or π depending on the sign of $\sin(\omega/2)$.

The sketches are given below:



Note that for the phase, the convention is to keep it in $[-\pi, \pi]$. The phase is equivalent when we add $2k\pi$ to it. Therefore, for the part that exceeds this range, we move it to $[-\pi, \pi]$ by adding $2k\pi$ where k is an integer. In this plot, when $\omega > \frac{\pi}{2}$, the phase $2\omega > \pi$, we subtract 2π to keep the phase in the range $[-\pi, \pi]$.

Grading: 12 points

- +6 each for correct magnitude/phase graph (other equivalent phase graph is fine)

5

Solution:

Remember that $\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$, we try to introduce a similar term in $H(\omega)$ by factoring out $e^{-j\frac{\omega}{2}}$:

$$H(\omega) = 1 - e^{-j\omega} = e^{-j\frac{\omega}{2}} \left(e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}} \right)$$

Notice that $(e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}) = 2j \sin\left(\frac{\omega}{2}\right)$. The expression becomes:

$$H(\omega) = 2j \sin\left(\frac{\omega}{2}\right) e^{-j\frac{\omega}{2}} = 2 \sin\left(\frac{\omega}{2}\right) e^{j(\frac{\pi}{2} - \frac{\omega}{2})}$$

Since this expression is in polar form, we can pull out the expressions for magnitude and phase:

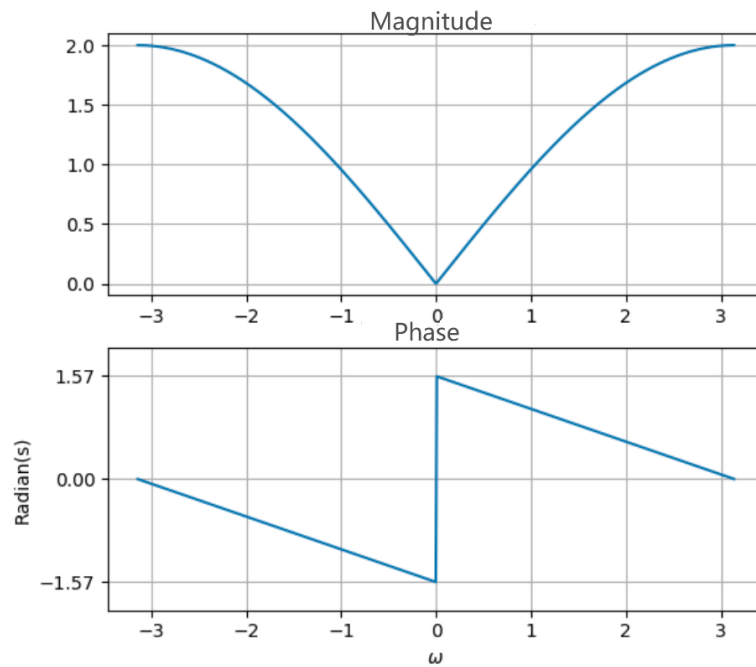
$$|H(\omega)| = 2 \left| \sin\left(\frac{\omega}{2}\right) \right|, \quad \angle H(\omega) = \frac{\pi}{2} - \frac{\omega}{2}$$

However, over the given interval, the sine term is negative when $\omega < 0$. Hence, we need to subtract π from the phase when $\omega < 0$. The magnitude and phase are given by:

$$|H(\omega)| = 2 \left| \sin\left(\frac{\omega}{2}\right) \right|, \quad (\text{Magnitude})$$

$$\angle H(\omega) = \begin{cases} \frac{\pi}{2} - \frac{\omega}{2}, & 0 < \omega \leq \pi \\ -\frac{\pi}{2} - \frac{\omega}{2}, & -\pi \leq \omega \leq 0 \end{cases} \quad (\text{Phase})$$

The sketches of the magnitude and phase are given below.

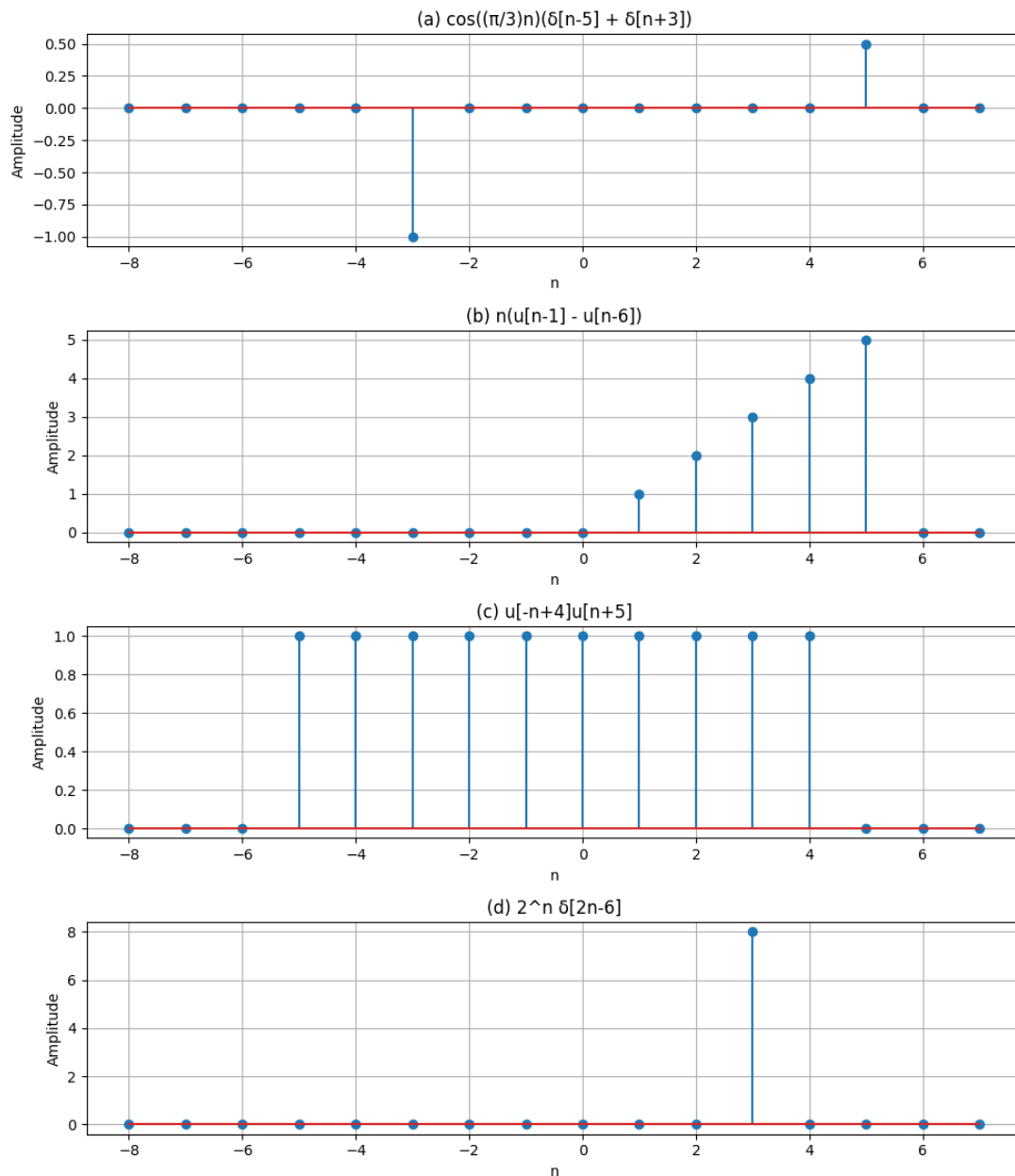


Grading: 12 points

- +4 for correct expression of magnitude (+2 if minor math errors)
- +4 for correct expression of phase (+2 if minor math errors)
- +2 each for correct magnitude/phase graph (other equivalent phase graph is fine)

Solution:

The sketches are shown below. Here are several remarks: (a) By definition of the Kronecker delta function, it only takes non-zero value when $n = 5$ and $n = -3$. (b) By definition of the unit step signal, the function $u[n] - u[n - 5]$ is only non-zero when $0 \leq n \leq 4$ where it takes a value of 1, and 0 otherwise. (c) Again, the function $u[-n + 5]u[n + 3]$ is non-zero only if both $u[-n + 5]$ and $u[n + 3]$ are non-zero, that corresponds to $n \leq 5$ and $n \geq -3$, respectively. (d) Similar to part (a), it is non-zero only if $2n = 6$, which is equivalent to $n = 3$.



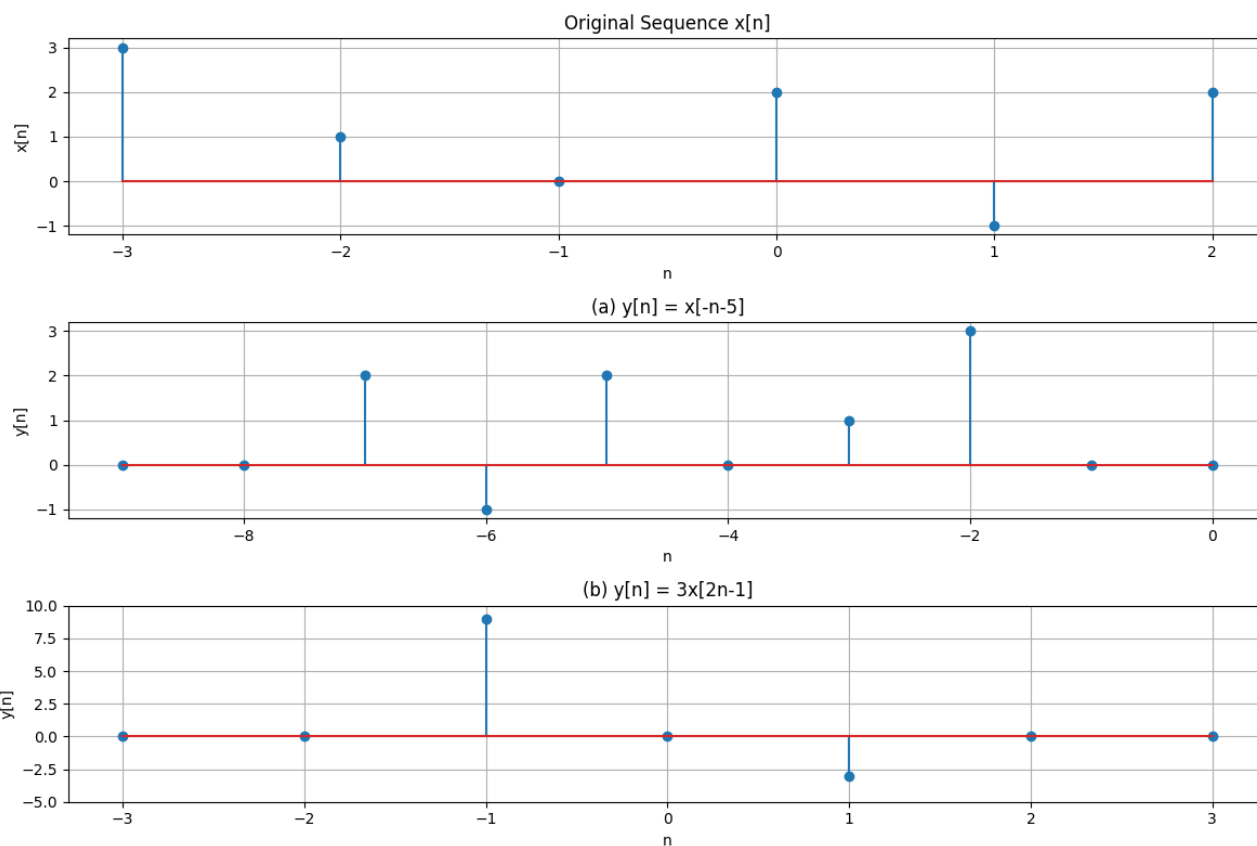
Grading: 12 points

- +3 for each correct sketch of (a), (b), (c) and (d)

Solution:

(a) One way to sketch $y[n]$ is to first shift $x[n]$ and then flip it around zero (or we can first flip then shift). To this end, let us define $z_1[n] = x[n - 5]$ be the shifted signal. Then we flip it by letting $y[n] = z_1[-n] = x[-n - 5]$.

(b) Similarly, we first shift and amplify $x[n]$, then scale it.



Grading: 12 points

- +6 (a) correct final sketch of $y[n]$
- +6 (b) correct final sketch of $y[n]$

Solution:

$$\{x[n]\} = \{\dots, 0, 1, 0, \underset{\uparrow}{3}, 0, -5, 1, \dots\}$$

$$\{x[n]\} = \delta[n+2] + 3\delta[n] - 5\delta[n-2] + \delta[n-3]$$

Grading: 12 points

I

- +3 each Delta